

<Limits>

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x}$$

$$\Rightarrow \frac{\ln(\infty)}{\infty} = \frac{\infty}{\infty} \quad \text{I.F.}$$

L.R:

$$\frac{(\ln x)'}{(x)'} = \frac{\frac{1}{x}}{1} = \frac{0}{1} = 0$$

$$\Rightarrow \frac{1}{\infty} = 0$$

$$\lim_{x \rightarrow 0^+} \frac{x \ln x}{1}$$

$$\Rightarrow \frac{\ln x}{\frac{1}{x}} \Rightarrow \frac{\ln(0)}{\frac{1}{0}} = \frac{0}{0} \quad \text{I.F.}$$

L.R:

$$\frac{(\ln x)'}{(\frac{1}{x})'} = \frac{\frac{1}{x}}{-\frac{1}{x^2}} = \frac{1}{x} \left(-\frac{x^2}{1} \right) = \frac{-x^2}{x} = -x$$
$$\Rightarrow \frac{x \ln x}{1} = -\odot$$

Done

~~E.F.~~